Practice Sets 1 and 2: Wrong-Answer Notes

CFA Level II · 15 wrong answers, one per page

40 / 48	41 / 48	81 / 96
Set 1 (83.3%)	Set 2 (85.4%)	Both sets (84.4%)

Score by topic (both sets)

Topic	Score
Equity Valuation	2/5
Derivatives	6/9
Financial Statement Analysis	11/13
Fixed Income	20/23
Ethical and Professional Standards	7/8
Portfolio Management	16/18
Quantitative Methods	15/16
Economics	3/3
Alternative Investments	1/1

Questions at a glance

Q	Topic	You	Ans
Set 1 Q16	FCFE: working capital excludes cash, and both investment terms use (1 − debt ratio)	C	B
Set 1 Q17	Model choice: FCFE when dividends do not reflect the ability to pay	B	A
Set 1 Q21	Standard VII(B): state the designation, never imply superiority or promise results	A	C
Set 1 Q30	Forward contract value after initiation: discount to today	A	C
Set 1 Q33	Accruals regression: the residual is the discretionary part	C	A
Set 1 Q35	Forward price of a zero-coupon bond: exponent = length of the forward period	A	C
Set 1 Q37	Turning fixed-rate debt into floating: receive fixed, pay floating	C	B
Set 1 Q42	DLOC and DLOM: convert the control premium, then multiply the discounts	B	A
Set 2 Q10	Style from active sensitivities (fund − benchmark): positive size = small cap, negative value = growth	B	A
Set 2 Q12	Callable bond when rates fall: duration falls and convexity turns negative	C	B
Set 2 Q16	Text cleansing removes noise; wrangling works on the tokens	B	A
Set 2 Q22	Calibrated tree: an option-free bond has the same value whatever the volatility	B	C
Set 2 Q29	Futures arbitrage: compare market price with the model price first	C	B
Set 2 Q34	Index fund that meets its objective: active return 0, so IR = 0	A	C
Set 2 Q42	Diluted shares: options AND RSUs use the treasury stock method with unrecognised expense	B	A

Patterns

- **Sign and direction:** swap to turn fixed debt into floating (Q37), buy or sell futures from the model price (Q29), active style from fund minus benchmark (Q10), callable duration when rates fall (Q12).
- **Steps skipped in formulas:** cash out of working capital and (1 − DR) on both terms (FCFE), discount the forward value to today, premium to discount before combining DLOC and DLOM, treasury stock method also for RSUs.
- **Definitions:** cleansing vs wrangling, discretionary = residual accruals, IR of an index fund = 0, option-free bond unaffected by volatility.

Set 1 Q16 FCFE: working capital excludes cash, and both investment terms use (1 − debt ratio)

You **C** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Tobias Weller values Grünwald AG with free cash flow models. Exhibit 1 shows data for the year (in € millions).

Exhibit 1: Grünwald AG Data

Item	Amount
Net income	420
Depreciation	150
Fixed capital investment	310
Increase in current assets (including a €30 million increase in cash)	110
Increase in current liabilities	40
Target debt ratio	40%

Question 16. Grünwald's FCFE, assuming investment is financed at the target debt ratio, is closest to:

OPTIONS

A	€260 million		€260m = 420 − (310 − 150): ignores the debt financing and the working capital investment.
B	€300 million	✓ Correct	Correct: 420 − 0.60 × 160 − 0.60 × 40 = €300m.
C	€236 million	✗ Your pick	Your pick: €236m = 420 − 160 − 24 takes the full net fixed investment without (1 − DR); only the working capital term was multiplied by 0.60.

Trigger "investment is financed at the target debt ratio"
"Increase in current assets (including a €30 million increase in cash)"

Rule next time: Cash is not working capital: take it out of the current assets increase. Then multiply BOTH the net fixed investment and the working capital investment by (1 − DR).

FORMULA / RULE

FCFE = NI − (1 − DR)(FCInv − Dep) − (1 − DR)(WCInv)
WCInv = Δ(current assets excluding cash) − Δ(current liabilities)

WHY THE ANSWER IS B

1. WCInv = (110 − 30) − 40 = 40.
2. Net fixed investment = 310 − 150 = 160.
3. FCFE = 420 − 0.60 × 160 − 0.60 × 40 = 420 − 96 − 24 = €300m: **B**.

Apply to other questions: Step 1: strip cash out of working capital. Step 2: apply (1 − DR) to every investment term.

Set 1 Q17 Model choice: FCFE when dividends do not reflect the ability to pay

You **B** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

Grünwald pays dividends that are far below its FCFE, and its dividends have little relation to its ability to pay.

Question 17. Weller should most likely value Grünwald with:

OPTIONS

A	an FCFE model, because dividends differ greatly from the capacity to pay them	✓ Correct	Correct: dividends differ greatly from the capacity to pay them, so FCFE is the right base.
B	a residual income model, because FCFE is positive	✗ Your pick	Your pick: a residual income model is chosen when FCFE is negative or there are no dividends or the cash flows are hard to forecast. A positive FCFE is not a reason for it.
C	a dividend discount model, because Grünwald pays dividends		A dividend discount model needs dividends that follow earnings. Paying dividends alone is not enough.

Trigger “dividends are far below its FCFE”
“little relation to its ability to pay”

Rule next time: Dividends far from FCFE and unrelated to capacity: use FCFE, not a dividend discount model.

FORMULA / RULE

DDM: stable dividends linked to earnings, minority view
FCFE: dividends differ from capacity to pay, or control perspective
Residual income: no dividends or negative free cash flow

WHY THE ANSWER IS A

1. The vignette says dividends are far below FCFE and not linked to ability to pay.
2. So the dividends do not measure the cash the company can distribute.
3. FCFE model: **A**.

Apply to other questions: Match the model to the clue words: “dividends differ from FCFE” = FCFE; “no dividends / negative FCF” = residual income.

Set 1 Q21 Standard VII(B): state the designation, never imply superiority or promise results

You **A** · Answer **C**

VIGNETTE (RELEVANT PART, UNEDITED)

Vass's brochure for new clients contains the statements listed in the last question of this vignette.

Question 21. Which statement in Vass's brochure complies with Standard VII(B): Reference to CFA Institute, the CFA Designation and the CFA Program?

OPTIONS

A	"As a CFA charterholder, Henrik Vass will earn you higher returns."	✗ Your pick	Your pick: promises higher returns because of the charter. That is a guarantee/superiority claim: not allowed.
B	"Henrik Vass passed all three CFA exams on his first attempt, which makes him a better analyst."		"Passed all three exams on first attempt, which makes him a better analyst" implies superior ability: not allowed.
C	"Henrik Vass is a CFA charterholder."	✓ Correct	Correct: "is a CFA charterholder" is a plain fact.

Trigger "Standard VII(B): Reference to CFA Institute, the CFA Designation and the CFA Program"

Rule next time: You may say you hold the designation. You may not link it to better returns or to being a better analyst.

FORMULA / RULE

> Allowed: "John Doe is a CFA charterholder"
> Not allowed: higher returns, better analyst, superior skill, or a promised result because of the CFA

WHY THE ANSWER IS C

1. Look at each statement: does it state a fact, or claim better results or ability?
2. Only statement C is a fact: **C**.

Apply to other questions: VII(B) trap words: "will earn you more", "better analyst", "superior".

Set 1 Q30 Forward contract value after initiation: discount to today

You **A** · Answer **C**

VIGNETTE (RELEVANT PART, UNEDITED)

Ten months ago, Marco Bellini entered a long 14-month forward contract on one share of Lumen Foods, a stock that pays no dividends, at a forward price of €88.40. Today Lumen's share price is €94.10, so four months remain until expiration. The risk-free rate is 2.2% a year with annual compounding.

Question 30. The current value of Bellini's forward position is closest to:

OPTIONS

A	€6.39	✗ Your pick	Your pick: €6.39 = 94.10 × 1.022^(4/12) − 88.40 is the value at expiration, not today. It is not discounted.
B	€5.70		€5.70 = 94.10 − 88.40 ignores the time value: F_0 must be discounted.
C	€6.34	✓ Correct	Correct: 94.10 − 88.40 ÷ 1.022^(4/12) = €6.34.

Trigger “entered a long 14-month forward contract … at a forward price of €88.40”
“four months remain until expiration”
“pays no dividends”

Rule next time: Long forward value = $S_0 - F_0 \div (1 + r)^{(T - t)}$, using the time left (4/12). The result is a value today, so the forward price is discounted.

FORMULA / RULE

> $V_t(\text{long}) = S_t - F_0 \div (1 + r)^{(T - t)}$ (no dividends)
> Same value: $(F_t - F_0) \div (1 + r)^{(T - t)}$, with $F_t = S_t \times (1 + r)^{(T - t)}$

WHY THE ANSWER IS C

1. T − t = 4/12 year. 1.022^(4/12) = 1.00729.
2. 88.40 ÷ 1.00729 = 87.77.
3. 94.10 − 87.77 = €6.34: **C**.

Apply to other questions: “Value now” always means discounting; use only the REMAINING time, not the original 14 months.

Set 1 Q33 Accruals regression: the residual is the discretionary part

You **C** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

Nora Lindahl screens companies for earnings quality.

Company R: a Beneish M-score of −1.2 and a high level of accruals. Lindahl separates accruals into normal accruals explained by a regression on business activity, and the residual.

Question 33. The residual from Lindahl's accruals regression represents:

OPTIONS

A	discretionary accruals, which may signal earnings manipulation	✓ Correct	Correct: the residual is discretionary accruals and may signal manipulation.
B	cash earnings		Cash earnings are the cash flow part of earnings, not accruals.
C	non-discretionary accruals, which arise from normal operations	✗ Your pick	Your pick: non-discretionary accruals are the part the regression EXPLAINS, not the residual.

Trigger “normal accruals explained by a regression on business activity, and the residual”

Rule next time: Explained by the regression = normal (non-discretionary). Residual = unusual (discretionary), the warning sign of manipulation.

FORMULA / RULE

Non-discretionary accruals = explained by business activity (normal)

Discretionary accruals = the residual (management choices, possible manipulation)

WHY THE ANSWER IS A

1. Regression explains normal accruals.
2. What is left over (the residual) is unusual: **A**.

Apply to other questions: Residual = “unexplained” = discretionary = suspicious.

Set 1 Q35 Forward price of a zero-coupon bond: exponent = length of the forward period

You **A** · Answer **C**

VIGNETTE (RELEVANT PART, UNEDITED)

Diego Navas, a fixed-income analyst, computes forward prices. The three-year forward rate two years from now, f(2,3), is 3.2% (annual compounding).

Question 35. The forward price, two years from now, of a zero-coupon bond paying 1 at year 5 is closest to:

OPTIONS

A	0.8543	✗ Your pick	Your pick: 0.8543 = 1 ÷ 1.032^5 uses j + k = 5 as the exponent.
B	0.9690		0.9690 = 1 ÷ 1.032^1 uses an exponent of 1.
C	0.9098	✓ Correct	Correct: 1 ÷ 1.032^3 = 0.9098.

Trigger “three-year forward rate two years from now, f(2,3)”
“zero-coupon bond paying 1 at year 5”

Rule next time: F(j,k) = 1 ÷ (1 + f(j,k))^k. The exponent is k, the 3 years the bond still has after the forward date.

FORMULA / RULE

> F(j,k) = 1 ÷ (1 + f(j,k))^k
> f(j,k) is quoted per year; k = years of the bond after the forward date

WHY THE ANSWER IS C

1. The bond is a 3-year zero starting two years from now.
2. 1.032^3 = 1.0990.
3. 1 ÷ 1.0990 = 0.9098: **C**.

Apply to other questions: Exponent = k (length of the forward period), never j + k.

Set 1 Q37 Turning fixed-rate debt into floating: receive fixed, pay floating

You **C** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Harlow Utilities has issued fixed-rate bonds. Its CFO, Lena Brandt, says: "I want our debt to behave like floating-rate debt so that the market value of our liabilities is less sensitive to changes in interest rates."

Question 37. To meet Brandt's objective, Harlow should enter into a swap in which it:

OPTIONS

A	pays fixed and receives an equity return		Pay fixed and receive an equity return adds more fixed payments and equity risk: does not give floating debt.
B	pays floating and receives fixed	✓ Correct	Correct: receive fixed, pay floating: fixed payments cancel, floating payments remain.
C	pays fixed and receives floating	✗ Your pick	Your pick: pay fixed, receive floating makes Harlow pay fixed twice. Its liabilities become MORE rate sensitive, the opposite of the goal.

Trigger “I want our debt to behave like floating-rate debt”
“fixed-rate bonds”

Rule next time: The bonds already make Harlow PAY fixed. To end up floating, the swap must RECEIVE fixed (cancels the bond coupon) and pay floating.

FORMULA / RULE

> Fixed-rate liability → floating: swap that receives fixed and pays floating
> Floating-rate liability → fixed: swap that pays fixed and receives floating

WHY THE ANSWER IS B

1. Existing position: pays fixed on bonds.
2. Swap receives fixed (offsets bond coupons) and pays floating.
3. **B**.

Apply to other questions: First write the cash flows you already pay, then choose the swap leg that cancels them.

Set 1 Q42 DLOC and DLOM: convert the control premium, then multiply the discounts

You **B** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

Weller values a small private company, Hallberg GmbH, for a minority investor. A controlling interest would command a control premium of 25%, and the discount for lack of marketability is 20%.

Question 42. The total discount applied to a controlling value to value Hallberg's minority, non-marketable stake is closest to:

OPTIONS

A	36%	✓ Correct	Correct: 1 − 0.80 × 0.80 = 36%.
B	40%	✗ Your pick	Your pick: 40% = 20% + 20% adds the two discounts. They multiply.
C	45%		45% = 25% + 20% adds and also uses the control premium as if it were DLOC.

Trigger “control premium of 25%”
“discount for lack of marketability is 20%”
Rule next time: The 25% is a control PREMIUM. Convert it: DLOC = 1 − 1/1.25 = 20%. Then total = 1 − (1 − DLOC)(1 − DLOM).

FORMULA / RULE

DLOC = 1 − 1/(1 + control premium)
Total discount = 1 − (1 − DLOC)(1 − DLOM)

WHY THE ANSWER IS A

1. DLOC = 1 − 1/1.25 = 20%.
2. Total = 1 − (1 − 0.20)(1 − 0.20) = 1 − 0.64 = 36%: **A**.

Apply to other questions: Premium → discount conversion first; never add two discounts.

Set 2 Q10 **Style from active sensitivities (fund − benchmark): positive size = small cap, negative value = growth** You **B** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

Ravi Kapoor evaluates the Orion Equity Fund against its benchmark using a three-factor model. Last year the fund returned 11.2% and the benchmark returned 8.1%.

Exhibit 1: Factor Sensitivities and Factor Returns

Factor	Fund sensitivity	Benchmark sensitivity	Factor return
Market	1.15	1.00	6.0%
Size	0.40	0.10	2.0%
Value	-0.10	0.20	4.0%

Question 10. Based on its sensitivities, the fund's style relative to the benchmark is best described as:

OPTIONS

A	high beta, small-cap and growth	✓ Correct	Correct: high beta, small-cap and growth.
B	high beta, large-cap and value	✗ Your pick	Your pick: high beta is right, but large-cap and value reverse the signs of the size and value differences.
C	low beta, large-cap and value		Low beta is wrong: 1.15 is above the benchmark's 1.00.

Trigger “Fund sensitivity | Benchmark sensitivity”
Size 0.40 vs 0.10; Value −0.10 vs 0.20
Rule next time: Style = fund sensitivity minus benchmark sensitivity: market +0.15 (high beta), size +0.30 (small cap), value −0.30 (growth).

FORMULA / RULE

> Active sensitivity = portfolio − benchmark
> + market = high beta; + size = small cap; + value = value; − value = growth; − momentum = contrarian

WHY THE ANSWER IS A

1. Market: 1.15 − 1.00 = +0.15.
2. Size: 0.40 − 0.10 = +0.30 → small cap.
3. Value: −0.10 − 0.20 = −0.30 → growth. **A**.

Apply to other questions: Subtract benchmark first, then read the signs.

Set 2 Q12 Callable bond when rates fall: duration falls and convexity turns negative

You **C** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Gabriela Silva, a portfolio manager at Tamsin Fixed Income, compares bonds with embedded options. All bonds have the same issuer, maturity and coupon unless stated otherwise.

Bond Q is callable, Bond R is putable and Bond S is an option-free (straight) bond.

Question 12. If interest rates fall sharply, compared with Bond S, Bond Q's effective duration most likely:

OPTIONS

A	rises and its convexity stays positive		Duration rising with positive convexity is the behaviour of the straight bond (and of a putable bond), not a callable.
B	falls and its convexity becomes negative	✓ Correct	Correct: duration falls and convexity becomes negative.
C	stays the same as Bond S's duration	✗ Your pick	Your pick: the callable matches the straight bond only when rates RISE (the call is far out of the money).

Trigger “Bond Q is callable”
“interest rates fall sharply”

Rule next time: Rates fall + callable = the call will be exercised: duration shortens, convexity turns negative (price compression).

FORMULA / RULE

Callable: rates fall → duration falls, negative convexity; rates rise → like straight

Putable: rates rise → duration falls; rates fall → like straight, positive convexity

WHY THE ANSWER IS B

1. Falling rates make the call likely.
2. The price is capped near the call price, so duration falls and convexity is negative.
3. **B**.

Apply to other questions: Callable is hurt when rates FALL; putable is protected when rates RISE.

Set 2 Q16 Text cleansing removes noise; wrangling works on the tokens

You **B** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

Arai's raw text contains HTML tags, punctuation, numbers and extra white space. He then applies four wrangling steps to the cleansed tokens: converting to lower case, removing words such as "the" and "is", reducing "integrated", "integrating" and "integration" to "integr", and converting "better" to "good" using a dictionary.

Question 16. Removing HTML tags, punctuation, numbers and white space is part of:

OPTIONS

A	text cleansing	✓ Correct	Correct: text cleansing.
B	text wrangling (preprocessing)	✗ Your pick	Your pick: wrangling (preprocessing) is the next step on the cleansed tokens (lowercase, stop words, stemming, lemmatisation).
C	text exploration		Exploration comes after wrangling (bag of words, DF, MI).

Trigger “raw text contains HTML tags, punctuation, numbers and extra white space”
“wrangling steps to the cleansed tokens”

Rule next time: Removing HTML tags, punctuation, numbers and white space = cleansing. Lowercasing, stop words, stemming and lemmatisation = wrangling.

FORMULA / RULE

Cleansing: remove HTML tags, punctuation, numbers, white space
Wrangling: lowercase, stop words, stemming, lemmatisation, n-grams
Exploration: bag of words, DF, MI

WHY THE ANSWER IS A

1. The listed items are noise in the raw text.
2. That is cleansing: **A**.

Apply to other questions: Ask: is it raw characters (cleansing) or tokens (wrangling)?

Set 2 Q22 Calibrated tree: an option-free bond has the same value whatever the volatility

You **B** · Answer **C**

VIGNETTE (RELEVANT PART, UNEDITED)

Bettina Kral, a fixed-income analyst, shows one interest rate path generated from a calibrated binomial tree. The tree has nodes at times 0, 1, 2 and 3. Along the path the one-year rates are 2.0% (time 0 to 1), 3.1% (time 1 to 2), 3.8% (time 2 to 3) and 4.4% (time 3 to 4).

She values a three-year, 4% annual-coupon, option-free bond along this path using the pathwise method.

Question 22. For an option-free bond, the value found by discounting at spot rates compared with the value from a calibrated binomial tree is:

OPTIONS

A	lower when the assumed volatility is higher		Volatility changing the value of an option-free bond is false: lower when volatility is higher is wrong.
B	higher when the assumed volatility is higher	✗ Your pick	Your pick: higher when volatility is higher is also wrong: no option, so no volatility effect.
C	the same, whatever volatility is assumed	✓ Correct	Correct: the same, whatever volatility is assumed.

Trigger “calibrated binomial tree”
“option-free bond”

Rule next time: A calibrated tree reprices the benchmark bonds, so an option-free bond is worth the same as with spot rates. Volatility only matters for embedded options.

FORMULA / RULE

> Option-free bond: tree value = spot-rate value
> Volatility ↑: callable value ↓, putable value ↑, straight value unchanged

WHY THE ANSWER IS C

1. The tree is built to match the benchmark curve.
2. Without options volatility has no effect.
3. **C**.

Apply to other questions: “Option-free” = volatility irrelevant.

Set 2 Q29 Futures arbitrage: compare market price with the model price first

You **C** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Ten-Year Note Futures

The underlying 2.5% semi-annual Treasury note has a dirty price of 105.20. It has been 40 days since the last coupon payment, and the futures contract expires in 90 days. No coupon is paid before expiration. The quoted futures price is 136.00, the annualised three-month risk-free rate is 1.80%, and the conversion factor is 0.7550.

Question 29. Based on the equilibrium price, the arbitrage strategy for the ten-year note futures is to:

OPTIONS

A	take no position, because the contract is fairly priced		Fairly priced is wrong: 136.00 differs from the model price.
B	buy the futures, short sell the note and lend the proceeds	✓ Correct	Correct: buy futures, short sell the note and lend the proceeds.
C	sell the futures, buy the note with borrowed funds	✗ Your pick	Your pick: sell futures, buy the note with borrowed funds is cash-and-carry, used when the futures is OVERPRICED.

Trigger “quoted futures price is 136.00”
“conversion factor is 0.7550”

Rule next time: Market 136.00 < model ≈ 138.76: the futures is cheap. Cheap futures = buy futures, short the note, lend (reverse cash-and-carry).

FORMULA / RULE

Model price = $[(S + AI_0)(1 + r)^T - AI_T - FV(\text{coupons})] \div CF$

Market > model: sell futures, buy bond, borrow

Market < model: buy futures, short bond, lend

WHY THE ANSWER IS B

1. Full price in 90 days: 105.20 × (1 + 0.018 × 90/360) = 105.67; subtract accrued at expiry 0.90 = 104.77.
2. Divide by 0.7550 = 138.76, above the market 136.00.
3. Futures underpriced → **B**.

Apply to other questions: Cheap futures = buy futures (and short the cash asset); expensive futures = sell futures.

Set 2 Q34 Index fund that meets its objective: active return 0, so IR = 0

You **A** · Answer **C**

VIGNETTE (RELEVANT PART, UNEDITED)

Felix Arnaud gives the investment committee three points on active management.

Point 1: An index fund that meets its objective exactly.

Question 34. The information ratio of the index fund in Point 1 is most likely:

OPTIONS

A	indeterminate but positive	✗ Your pick	Indeterminate but positive is wrong: active return is zero.
B	equal to its Sharpe ratio		Equal to its Sharpe ratio is wrong: IR uses active return over active risk, Sharpe uses excess return over the risk-free rate.
C	zero	✓ Correct	Correct: zero.

Trigger "An index fund that meets its objective exactly"

Rule next time: Meets its objective exactly = tracks the benchmark: active return 0, so the information ratio is 0.

FORMULA / RULE

IR = active return ÷ active risk

Index fund meeting its objective: IR = 0

WHY THE ANSWER IS C

1. Active return = fund return − benchmark return = 0.
2. IR = 0: **C**.

Apply to other questions: IR is always measured against the benchmark.

Set 2 Q42 Diluted shares: options AND RSUs use the treasury stock method with unrecognised expense

You **B** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

Aalto computes Kiruna's diluted shares. Basic weighted-average shares are 5,000,000 and the average share price is $30.

Exhibit 1: Potentially Dilutive Securities

Security	Number	Exercise price	Average unrecognised compensation expense
Stock options	200,000	$18	$600,000
Unvested RSUs	60,000	–	$900,000

Average unrecognised expense is the average of the beginning and ending unrecognised compensation expense.

Question 42. Kiruna's diluted shares are closest to:

OPTIONS

A	5,090,000	✓ Correct	Correct: 5,000,000 + 60,000 + 30,000 = 5,090,000.
B	5,120,000	✗ Your pick	Your pick: 5,120,000 adds all 60,000 RSUs without applying the treasury stock method.
C	5,260,000		5,260,000 adds all 200,000 options and all 60,000 RSUs and ignores the proceeds.

Trigger "Average unrecognised compensation expense" in the table for BOTH securities

Rule next time: RSUs have no exercise price, but the unrecognised expense still counts as proceeds: dilutive shares = 60,000 − 900,000 ÷ 30 = 30,000.

FORMULA / RULE

> Dilutive shares = number − (exercise proceeds + average unrecognised expense) ÷ average price

WHY THE ANSWER IS A

1. Options: 200,000 − (3,600,000 + 600,000) ÷ 30 = 60,000.
2. RSUs: 60,000 − 900,000 ÷ 30 = 30,000.
3. Diluted = 5,090,000: **A**.

Apply to other questions: RSUs: proceeds = 0 but the unrecognised expense is still the "proceeds".